The equation of a circle [Cambridge revision]

Centre and radius from the equation, and the equation from the centre and radius.

LESSON 55–56 | 2 × 40 MIN | REVISION OF GRADE 9 | P1 3.4

LESSON CLOCK

15' | 25' | 20' | 25' | 15'

The standard form	15 min
Completing the square	25 min
Inside, on or outside	20 min
Tangents	25 min
Homework	15 min

LEARNING OBJECTIVES

- Write the equation of a circle from its centre and radius.
- Complete the square to read the centre and radius from a general equation.
- Decide whether a point lies inside, on or outside a circle.
- Use the property that the tangent is perpendicular to the radius.

TEXTBOOK

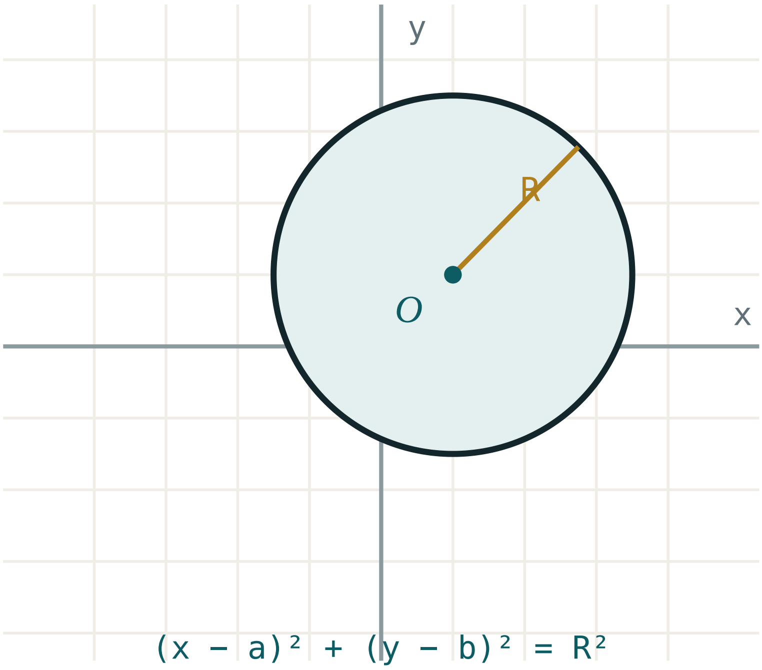
National	Geometry 9, pp. 119–132
Cambridge	Pure Mathematics 1, pp. 57–64
Workbook	Exercise 3D

01 Explanation

The standard form

A circle is the set of points at a fixed distance from a fixed point. Apply the distance formula and square:

$$(x - a)^2 + (y - b)^2 = r^2$$



Centre (a, b), radius r — the distance formula with the square roots removed.

CENTRE	R	EQUATION
$(0, 0)$	5	$x^2 + y^2 = 25$
$(3, -2)$	4	$(x - 3)^2 + (y + 2)^2 = 16$
$(-1, 0)$	$\sqrt{7}$	$(x + 1)^2 + y^2 = 7$

THE SIGNS INSIDE THE BRACKETS ARE REVERSED

A centre at $(3, -2)$ gives $(x - 3)^2$ and $(y + 2)^2$. Reading the centre straight off the brackets without changing the sign is the standard error in this topic.

Completing the square

Expanded, the equation looks unlike a circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Complete the square in x and in y separately to recover the standard form.

Example. $x^2 + y^2 - 6x + 4y - 12 = 0$:

STEP	WORKING
group	$(x^2 - 6x) + (y^2 + 4y) = 12$
complete	$(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$
tidy	$(x - 3)^2 + (y + 2)^2 = 25$
read	centre $(3, -2)$, radius 5

WHEN IT IS NOT A CIRCLE

If the right-hand side comes out **negative**, no point satisfies the equation; if it is **zero**, the “circle” is the single point at the centre. Both appear in examinations, and both want a sentence, not a radius.

Inside, on or outside

Compute the distance from the point to the centre and compare with r.

$$d < r \text{ inside, } d = r \text{ on, } d > r \text{ outside}$$

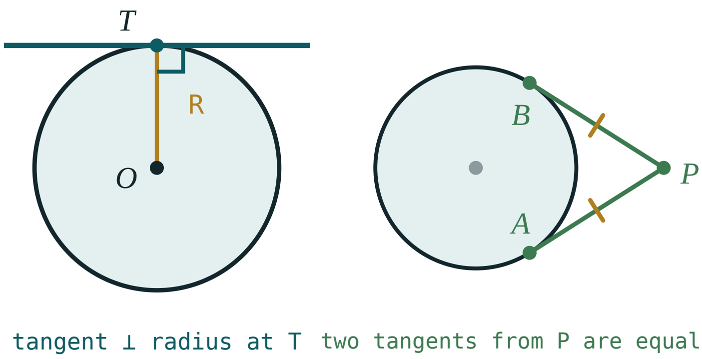
Faster still: substitute the point into $(x - a)^2 + (y - b)^2 - r^2$ and read the sign — negative inside, zero on, positive outside. No square root is needed.

POINT	SUBSTITUTED INTO $(X-3)^2 + (Y+2)^2 - 25$	WHERE
$(3, 3)$	$0 + 25 - 25 = 0$	on
$(5, 0)$	$4 + 4 - 25 = -17$	inside
$(10, 2)$	$49 + 16 - 25 = 40$	outside

Tangents

THE ONE CIRCLE PROPERTY THIS TOPIC NEEDS

The tangent at a point is **perpendicular to the radius** drawn to that point.



Radius and tangent meet at a right angle — the whole of the tangent method.

So the tangent at P on a circle with centre C is found in two steps: compute m_{CP}, then use $-\frac{1}{m_{CP}}$ through P.

Example. The tangent to $x^2 + y^2 = 25$ at $(3, 4)$. Here $m_{CP} = \frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:

$$y - 4 = -\frac{3}{4}(x - 3), \text{ that is } 3x + 4y = 25$$

A PATTERN WORTH REMEMBERING

For the circle $x^2 + y^2 = r^2$ the tangent at (x_0, y_0) is simply $x_0x + y_0y = r^2$. Check: at $(3, 4)$ on $x^2 + y^2 = 25$ it gives $3x + 4y = 25$ at once.

02 Worked examples

EXAMPLE 1 **Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.**

1. $(x^2 - 6x) + (y^2 + 4y) = 12$
2. $(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$
3. $(x - 3)^2 + (y + 2)^2 = 25$
4. Centre $(3, -2)$, radius 5.

ANSWER

Centre $(3, -2)$, radius 5

EXAMPLE 2 **Find the equation of the circle with $A(1, 2)$ and $B(7, 10)$ as ends of a diameter.**

1. Centre = midpoint = $(4, 6)$.
2. $AB = \sqrt{36 + 64} = 10 \Rightarrow r = 5$
3. $(x - 4)^2 + (y - 6)^2 = 25$

ANSWER

$(x - 4)^2 + (y - 6)^2 = 25$

EXAMPLE 3 **Where does $(6, 1)$ lie relative to $(x - 3)^2 + (y + 2)^2 = 25$?**

1. $(6 - 3)^2 + (1 + 2)^2 = 9 + 9 = 18$
2. $18 < 25$
3. Inside.
   No square root needed.

ANSWER

Inside the circle

EXAMPLE 4 **Find the tangent to $x^2 + y^2 = 25$ at $(-4, 3)$.**

1. $m\_\{CP\} = -\dfrac{3}{4}$
2. $m\_\{tangent\} = \dfrac{4}{3}$
3. $y - 3 = \dfrac{4}{3}(x + 4)$
4. $4x - 3y + 25 = 0$
   Or use $x_0x + y_0y = 25$ directly.

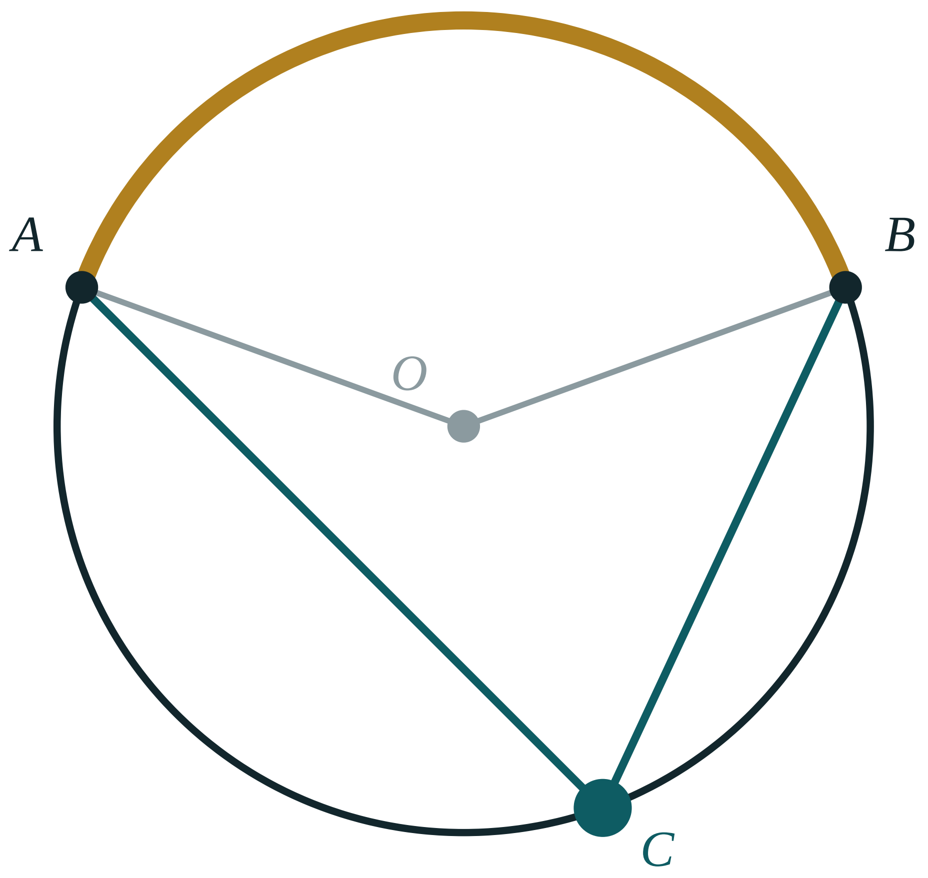
ANSWER

$4x - 3y + 25 = 0$

03 Interactive model

Complete the square on the board twice — once with a positive and once with a negative constant — and let the class say what the second one is.

The circle and its lines



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Centre	Markaz	Центр
Radius	Radius	Радиус
Standard form	Kanonik ko‘rinish	Каноническое уравнение
General form	Umumiy ko‘rinish	Общее уравнение
Completing the square	To‘la kvadratga keltirish	Выделение полного квадрата
Tangent	Urinma	Касательная
Chord	Vatar	Хорда
Diameter	Diametr	Диаметр
Concentric	Konsentrik	Концентрические

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The circle $(x - 3)^2 + (y + 2)^2 = 16$ has centre:

- $(3, 2)$
- $(3, -2)$
- $(-3, 2)$
- $(-3, -2)$

Its radius is:

- 16
- 4
- 8
- $\sqrt{16}^{2}$

To find the centre from the general form you:

- factorise
- complete the square
- differentiate
- substitute

If the right side comes out negative, the equation gives:

- a small circle
- no points at all
- a point
- a line

A tangent meets the radius at:

- $45°$
- $90°$
- $60°$
- any angle

The tangent to $x^2 + y^2 = 25$ at $(3, 4)$ is:

- $3x + 4y = 25$
- $4x + 3y = 25$
- $3x - 4y = 25$
- $x + y = 7$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Equation of the circle, centre $(0, 0)$, radius 5

   ANSWER $x^2 + y^2 = 25$

2. Equation, centre $(3, -2)$, radius 4

   ANSWER $(x - 3)^2 + (y + 2)^2 = 16$

3. Centre of $(x + 1)^2 + (y - 5)^2 = 9$

   ANSWER $(-1, 5)$

4. Radius of $(x + 1)^2 + (y - 5)^2 = 9$

   ANSWER 3

5. Radius of $x^2 + y^2 = 20$

   ANSWER $2\sqrt{5}$

6. Does $(3, 4)$ lie on $x^2 + y^2 = 25$?

   ANSWER Yes

7. Centre of $x^2 + y^2 = 49$

   ANSWER $(0, 0)$

Medium · the standard question

7 PROBLEMS

1. Centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$

   ANSWER $(3, -2), 5$

2. Centre and radius of $x^2 + y^2 + 8x - 2y + 8 = 0$

   ANSWER $(-4, 1), 3$

3. Circle on the diameter $(1, 2)$ to $(7, 10)$

   ANSWER $(x - 4)^2 + (y - 6)^2 = 25$

4. Is $(6, 1)$ inside $(x - 3)^2 + (y + 2)^2 = 25$?

   ANSWER Yes — $18 < 25$

5. Tangent to $x^2 + y^2 = 25$ at $(3, 4)$

   ANSWER $3x + 4y = 25$

6. Tangent to $x^2 + y^2 = 25$ at $(-4, 3)$

   ANSWER $4x - 3y + 25 = 0$

7. Circle with centre $(2, 3)$ passing through $(5, 7)$

   ANSWER $(x - 2)^2 + (y - 3)^2 = 25$

Hard · stretch and reason

7 PROBLEMS

1. Show that $x^2 + y^2 - 4x + 6y + 20 = 0$ has no points

   ANSWER $r^2 = -7$

2. Circle through $(0,0), (6,0), (0,8)$

   ANSWER $(x - 3)^2 + (y - 4)^2 = 25$

3. Length of the tangent from $(10, 2)$ to $(x - 3)^2 + (y + 2)^2 = 25$

   ANSWER $\sqrt{40} \approx 6.32$

4. The circles $x^2 + y^2 = 4$ and $(x - 5)^2 + y^2 = 9$: do they touch?

   ANSWER Yes, externally — $5 = 2 + 3$

5. Shortest distance from $(10, 2)$ to that circle

   ANSWER $\sqrt{65} - 5 \approx 3.06$

6. Circle with centre on $y = x$, radius 5, through $(1, 1)$

   ANSWER $(x - a)^2 + (y - a)^2 = 25$ with $a = 1 \pm \dfrac{5}{\sqrt{2}}$

7. Equation of the circle inscribed in the square with vertices $(0,0), (6,0), (6,6), (0,6)$

   ANSWER $(x - 3)^2 + (y - 3)^2 = 9$

07 Homework

Homework — 5 tasks

Show the completing-the-square step in full; the answer alone earns half the marks.

1. Find the centre and radius of $x^2 + y^2 - 10x + 6y + 9 = 0$.
2. Find the equation of the circle with $(-2, 1)$ and $(6, 7)$ as ends of a diameter.
3. Decide whether $(1, 1)$, $(5, -3)$ and $(9, 4)$ lie inside, on or outside $(x - 5)^2 + (y + 3)^2 = 16$.
4. Find the tangent to $x^2 + y^2 = 100$ at $(6, 8)$, and check it with $x_0x + y_0y = r^2$.

5. Explain in two sentences why $x^2 + y^2 + 2x + 2y + 5 = 0$ has no solutions.